Ex No: 14

PACKET SNIFFING USING RAW SOCKETS

AIM:
To write a program using Raw Sockets to implement Packet Sniffing.

ALGORITHM:
1 Create a raw socket.
2 Bind the socket to the network interface
3 Recieve network packets as
4 Extract the packet header information
5 Display the source IP, destination IP and protocol
6 Continue capturing packets and display the details

CODE:

```
import socket
import struct
s = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
print("packet sniffer started ...")
while True:
    packet, addr = s.recvfrom(65565)
    IP-header = packet[0:20]
    ph = struct.unpack("!BBHHBBH4s4s", ip_header)
    source_ip = socket.inet_ntoa(iph[8])
    destination_ip = socket.inet_ntoa(iph[4])
    protocol = iph[6]
    print("Source IP:", source_ip)
    print("Destination IP:", destination_ip)
    print("protocol:", protocol)
    print("." * 40)
```

Result

Thus the program to implement packet striffing using RAW sockets was executed successfully